

Data Analysis Internship

Task 3 Report

Employee Performance, Pay, Satisfaction
and Attrition

Analysis carried out on

THE IBM HR ATTRITION DATASET

1,470 employee records

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1 Executive summary

The company loses 16.1% of its workforce, 237 employees of 1,470. That number is almost useless on its own. It ranges from 7.3% to 67.8% depending on three questions about an employee.

The file arrives with zero missing values, zero duplicate records, and it passes all eight internal consistency rules. It is still not ready to analyse. Three defects were found, and none of them is a missing value.

1. Three columns carry no information. `EmployeeCount`, `Over18` and `StandardHours` hold one value on every row.
2. The performance scale is censored. `PerformanceRating` contains only the values 3 and 4: not one employee in 1,470 is rated below 3. The performance analysis the brief asks for cannot be carried out, and this report says so rather than producing one built on an unusable column.
3. Three pay columns are not pay. `MonthlyRate`, `DailyRate` and `HourlyRate` correlate with actual monthly income at r below 0.04 in absolute value.

No row was deleted.

2 Key performance indicators

Table 1. Workforce indicators

Indicator	Value	Why it is on this list
Headcount	1,470	The denominator for everything below
Attrition rate	16.1%	The number the board asks for
Employees who left	237	The absolute cost behind the rate
Median monthly income	4,919	Median, not mean: income is right-skewed
Average tenure	7.0 years	Short tenure and high attrition are one problem seen twice
Average age	36.9 years	Frames the tenure figure
Share on overtime	28.3%	The strongest single lever

3 Method

Every figure in this report is an attrition rate within a segment, expressed as a lift against the company baseline:

$$\text{lift} = \frac{\text{attrition rate in the segment}}{\text{attrition rate in the company}} \quad (1)$$

A rate on its own is not comparable between segments: twenty per cent is alarming in one company and unremarkable in another. A lift of 2.0 means the segment leaves twice as often as the company average, and that statement travels.

Headcount is reported beside every rate, and segments below twenty employees are excluded from the ranked tables. A rate of fifty per cent computed on four people is not a finding.

4 Findings

4.1 Who leaves

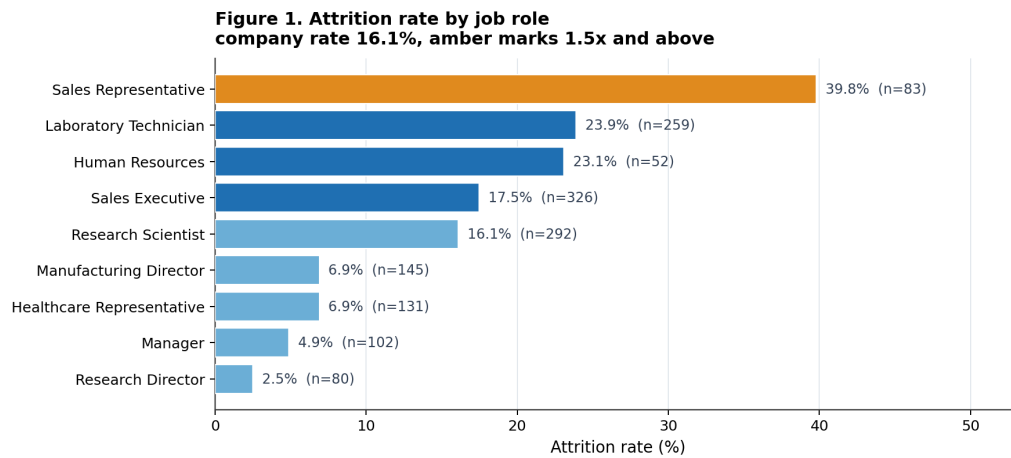


Figure 1: Attrition by job role. The dashed line is the company rate; a bar without its reference is not interpretable. Amber marks any segment at 1.5 times the company rate or above.

Sales Representative leave at 39.8%, a lift of 2.47.

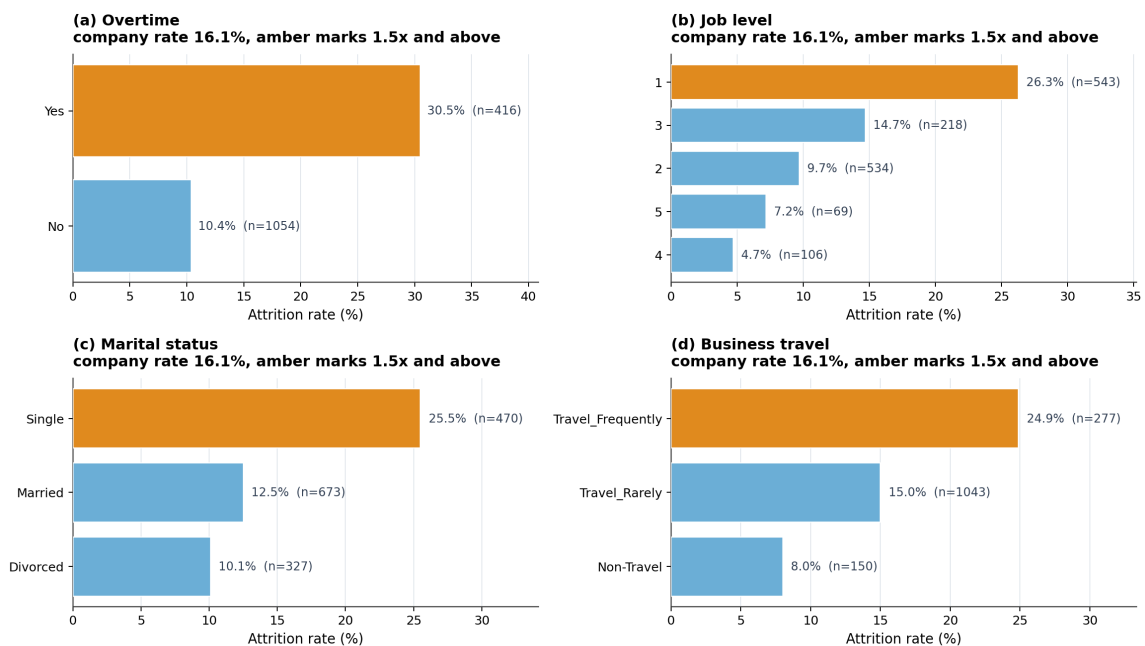


Figure 2: The four strongest single factors, on a shared scale. The shared scale is what makes the panels comparable.

4.2 Tenure

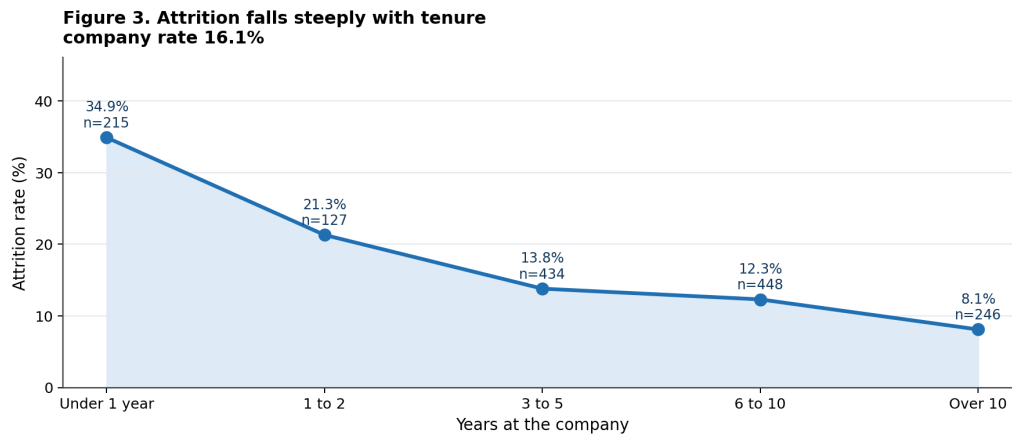


Figure 3: Attrition against tenure. A line rather than bars, because tenure is a progression and not a set of independent categories.

Employees with two years or less are **23.3% of the workforce** and **43.0% of all departures**. In the first year alone the rate reaches 34.9%.

4.3 Pay

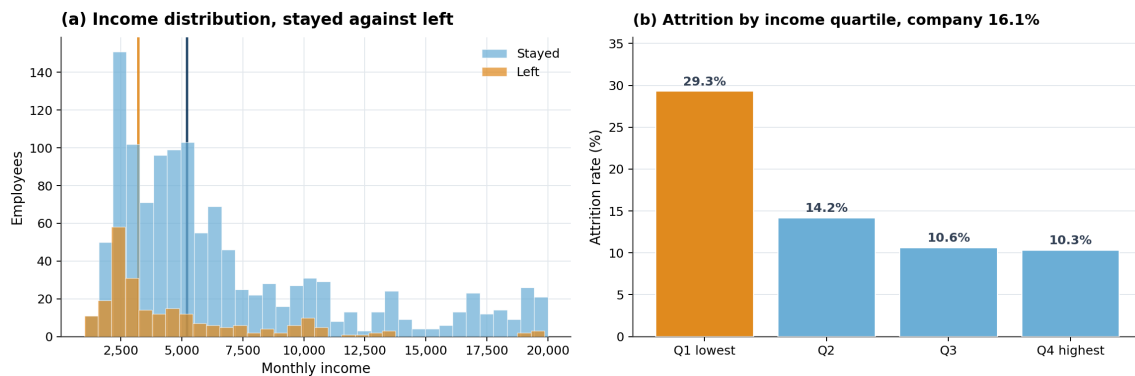


Figure 4: Pay and attrition. Panel (a) gives the distribution, panel (b) the rate by quartile. Neither alone is enough: the distribution shows where people sit, the rate shows where they leave.

4.4 Satisfaction

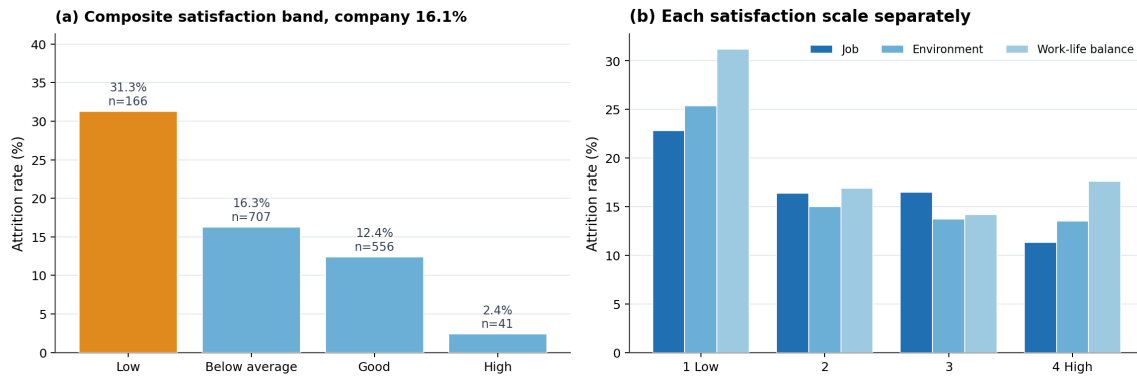


Figure 5: Satisfaction and attrition. Panel (b) places the three scales on one axis so their gradients can be compared directly.

4.5 The compounding effect

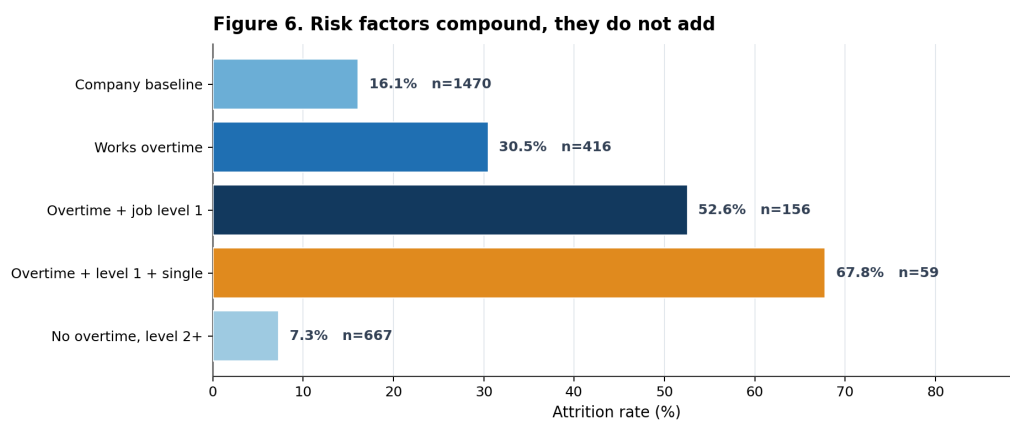


Figure 6: Risk factors compound rather than add. The progression from the company rate to 67.8% is the operational result of this study.

4.6 Power BI dashboard

The findings above are also packaged as an interactive Power BI dashboard, so that any of the segments described in this report can be explored directly, filtered by department and gender.

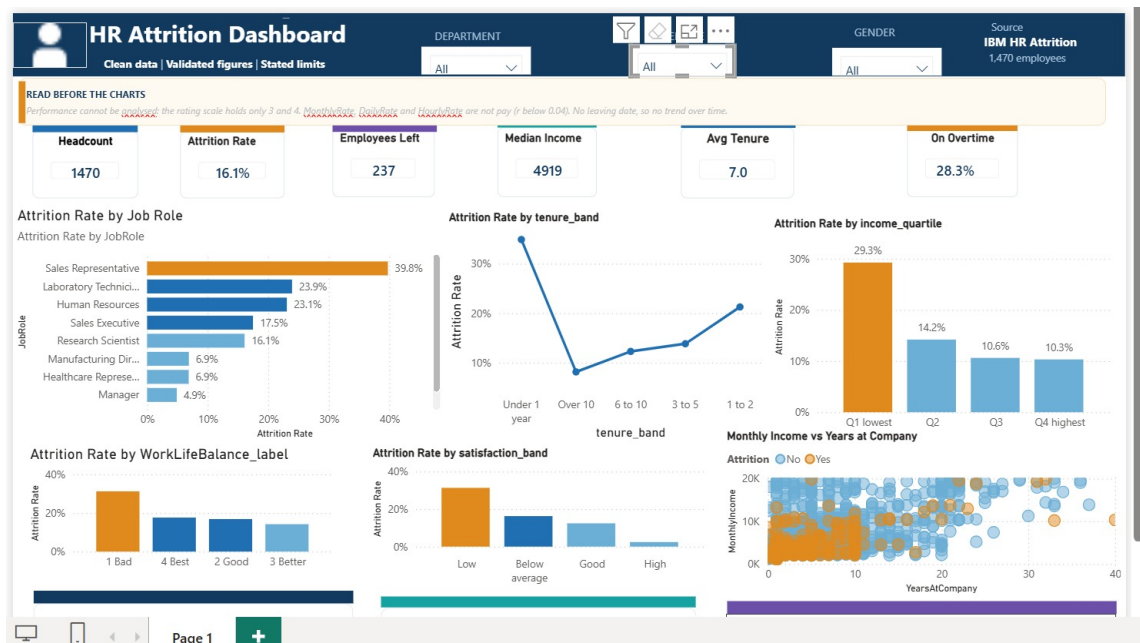


Figure 7: Power BI dashboard – overview page. KPI cards, attrition by job role, tenure band, income quartile, work–life balance and satisfaction band, and monthly income against years at the company.



Figure 8: Power BI dashboard – key insights, recommendations and data-quality panel, reproducing the conclusions of Sections 5 and 6 alongside the charts.

5 Insights and recommendations

5.1 Overtime is the strongest single predictor, and it is a management choice

Employees working overtime leave at 30.5%, those who do not at 10.4%, a 2.93-fold difference and the largest produced by any single field.

Its nature matters more than its size. Age, marital status and job level are attributes of the employee. Overtime is an outcome of how work is allocated, which makes it the one strong factor the company directly controls.

Recommendation. Audit overtime by manager and by role before touching pay or benefits. The lever is already in the company's hands.

5.2 Risk factors compound, they do not add

Group	Headcount	Attrition
Company baseline	1,470	16.1%
Works overtime	416	30.5%
Overtime and job level 1	156	52.6%
Overtime, level 1 and single	59	67.8%
No overtime, level 2 and above	667	7.3%

Recommendation. Target the 59 identified individuals rather than launching a general engagement programme, which spends most of its budget on the 667 employees leaving at 7.3%.

5.3 The first two years carry nearly half the departures

Recommendation. Move retention spending into the first 24 months: structured onboarding, an assigned mentor, a review at month 6 and month 18. An intervention at year five reaches a population already leaving at half the company rate.

5.4 Pay is a floor, not a ladder

Median income among leavers is 3,202 against 5,204 among those who stay. The effect is not linear: the lowest quartile leaves at 29.3%, the other three sit between 10.3% and 14.2% and barely differ.

Recommendation. Review the lowest quartile against the local market. Do not fund a general pay rise; it would change almost nothing above the first quartile.

5.5 Satisfaction matters, and performance cannot be measured

The lowest composite satisfaction band leaves at 31.3% against 2.4% in the highest. Work–life balance is the sharpest single scale, and it connects directly to the overtime finding: they are likely the same problem.

Against this, PerformanceRating correlates with income at -0.017 and contains only two of its four values.

Recommendation. Act on work–life balance, and rebuild the appraisal process. A scale on which nobody scores below 3 is not measuring performance.

6 Data quality

Table 2. Checks that passed

Check	Result
Missing values, all columns	0
Duplicate rows	0
Duplicate employee identifier	0
Years at company exceeds total working years	0
Last promotion predates the hire	0
Career implied to start before age 16	0
Satisfaction code outside 1 to 4	0

Constant columns. EmployeeCount, Over18 and StandardHours hold one value on every row and were dropped. Left in place they inflate a correlation matrix, and StandardHours = 80 invites the false conclusion that everyone works the same hours, which the overtime column contradicts for 28.3% of staff.

The censored performance scale. 1,244 employees are rated 3 and 226 are rated 4. The column is kept in the table and excluded from every conclusion, with the reason stated rather than the column quietly ignored.

The rate columns. Their correlations with monthly income are +0.035, +0.008 and −0.016. They are excluded from every pay figure. An analyst taking them at face value produces a pay analysis that is entirely noise, and nothing in the column name warns them.

7 Limitations

The data is synthetic. IBM generated it to demonstrate attrition analysis. Every finding here should be treated as a method demonstration until reproduced on the company's own records.

No leaving date. Attrition is a flag with no date, so no trend over time exists and a resignation cannot be separated from a dismissal.

No manager identifier. The overtime finding points at how work is allocated but cannot say by whom. This is the single most valuable field to add.

Correlation, not causation. That overtime accompanies departure does not prove it causes it. Testing that requires a controlled change, not more analysis of this file.

Deliverables

File	Content
Task3_HR_Analysis.ipynb	Colab-ready notebook, executed
report.pdf	The present report
KOUAME_Koffi_Fidèle_HR_Dashboard.pbix	Interactive Power BI dashboard
data/PowerBI_import.csv	Analysis table, no transformation required
data/HR_Attrition_clean.csv	Cleaned dataset
INSIGHTS.md, DATA_QUALITY.md	Conclusions and full audit
charts/	Six figures

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